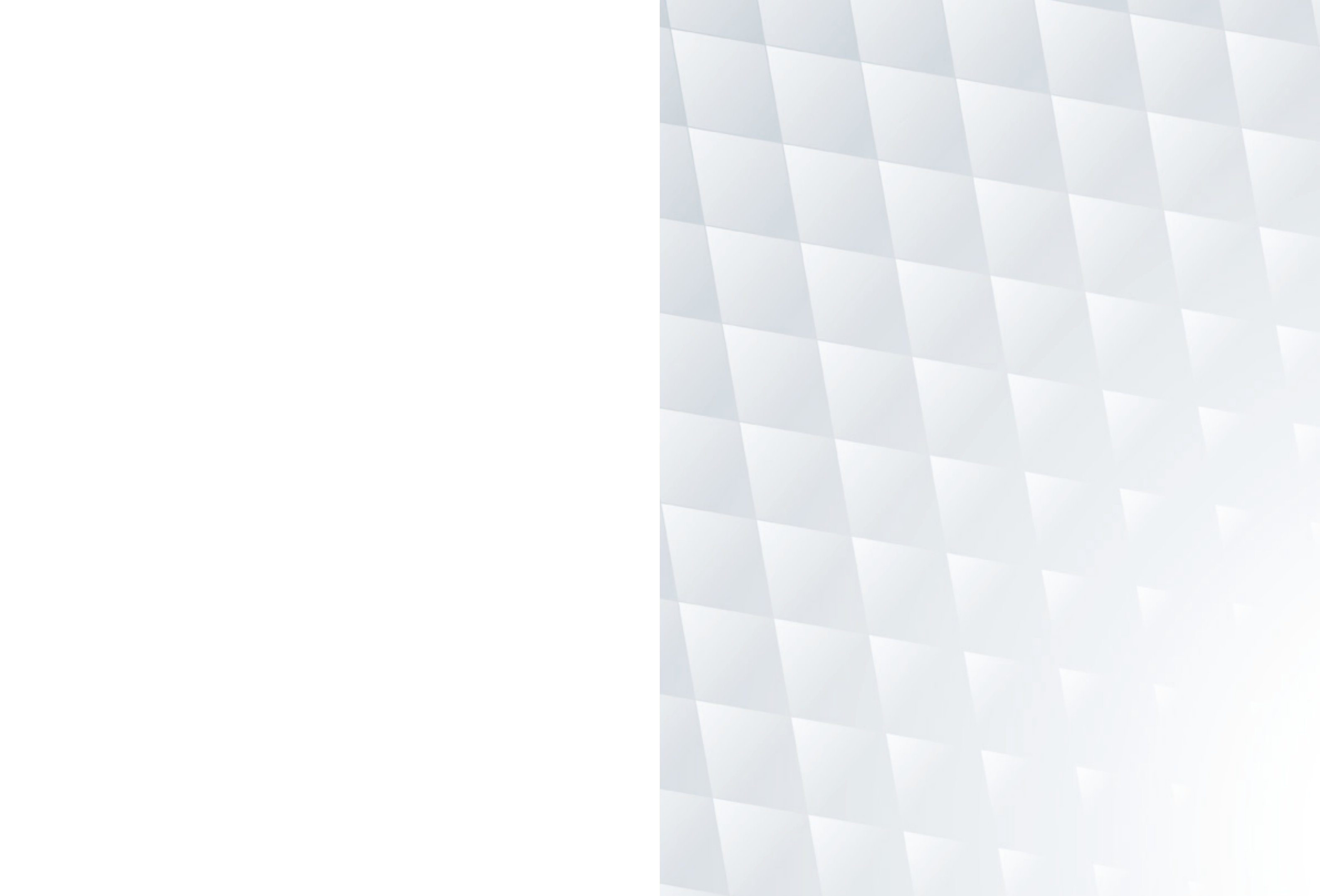


Romulus

Backward Curved Centrifugal Fan





Introduction of SEEMTEK

Mission

Committed to providing the most tailored fan solutions and products.

Vision

To be the leading brand in external rotor fan technology.

Value

Reliability, Value, Innovation.



Backward Centrifugal Fan of Romulus

Seemtek

Seemtek is derived from 'seem' and 'tech,' suggesting the company's philosophy of making high-tech solutions appear simple and intuitive. Just as the name implies, Seemtek designs fans and motors that blend seamlessly into various environments while delivering advanced performance.

These products, like the EC fans, embody the balance of simplicity and technology, providing energy-efficient solutions that adapt effortlessly to different applications, from HVAC systems to industrial use.

Our Slogan

We don't just provide products, our goal is to achieve our customers' objectives by offering comprehensive services, from Fan system design to Product delivery and Service. By helping customer engineers solve problems and enhance product performance, Seemtek creates value for its clients. This approach aligns perfectly with our mission: Tailoring Solutions for Fan Industry.

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01 Product Features



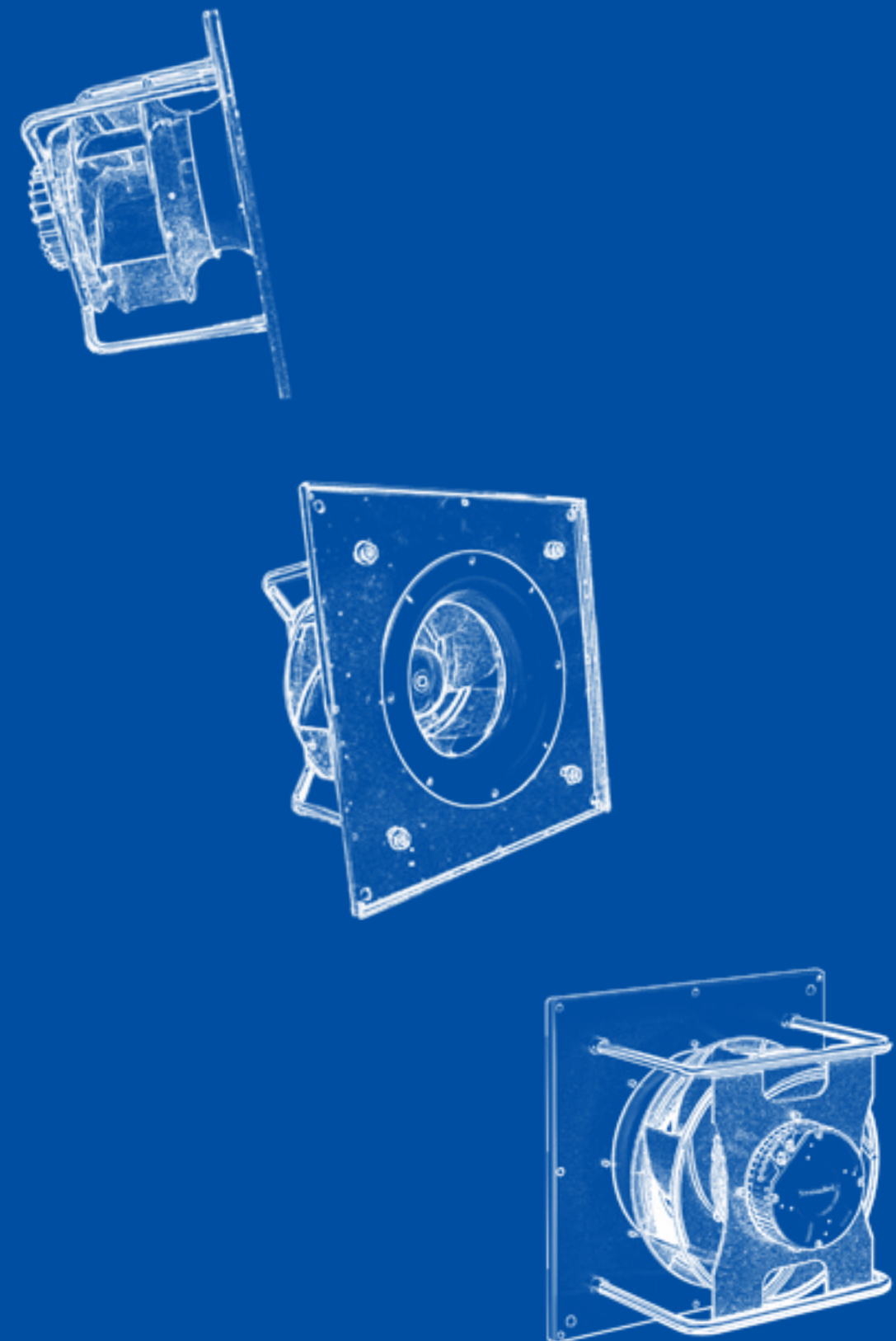
*Introduction of Each Part

*Comprehensive Features

*Comparative Advantage

Based on Real Customer Feedback

All of our features are gathered from the pain points faced by our customers' engineers when using fans. Our product engineers transform these pain points into specific technical requirements, which our R&D engineers then work to meet one by one during product development. This process shapes the unique features of our products.



Introduction of Each Part

Impeller

7-blade airfoil impeller with prime number asymmetric design greatly reduces high-speed airflow impact force, effectively dispersing energy loss of the fan and enhancing overall operational efficiency.

39° outlet installation angle, *15°-20° lower cover plate design ensures high static pressure while maintaining efficient fan operation, smooth transition of airflow pressure, and improved static pressure efficiency.

Integral injection molding structure design effectively controls radial and axial runout offset, greatly reducing motor operational losses.



Inlet Ring

Optimized diffuser design with best matching clearance and depth to enhance static pressure efficiency.

Gradual change in airflow direction through expanded orifices for uniform airflow organization.



Bracket



Integral installation for convenience.
High-strength design.
Resonance elimination design.

Control



Motor control mode is mainly based on advanced electronic commutation technology and intelligent control strategies to achieve efficient, precise, and reliable operation.

Supports FOC, PWM, power closed-loop, and torque control, and remote control via RS485, MODBUS, and other programs.

In FOC algorithm drive, the direction of the stator and rotor magnetic fields is maintained at a certain angle (e.g., 90°) through precise control, achieving maximum torque output under given current.

Motor



Utilizes permanent magnet excitation to eliminate losses from induction motor excitation current, improving motor efficiency. Brushless design provides high torque and speed to meet various high load and high-performance requirements.

Sine wave output with smooth waveform helps reduce noise and vibration during motor operation, reduces electromagnetic interference, and improves operational efficiency and control accuracy.

Uses ceramic shaft material to prevent bearing electro-corrosion.

Comprehensive Features

Romulus Centrifugal Fan

An ideal solution for the new generation FFUs, ventilation and air conditioning systems, data centers, and other fields. The Romulus backward-inclined centrifugal fan is the latest high-efficiency centrifugal fan series introduced by Seemtek. The motor boasts compactness, high system efficiency, and high power density.

Simultaneously, the outer rotor structure enhances driving torque, and the newly developed aerodynamic impeller for this motor can provide high efficiency even at full load or under partial loads.

Romulus is equipped with commonly used signal input and output interfaces in the industry, enabling smart control through Modbus communication.

Furthermore, the professional design of this series allows for a wider efficient operating range, while achieving high power density and low noise levels at operating points.

High Efficiency

The clever combination of the motor and impeller allows for unobstructed airflow into the impeller, achieving high transmission efficiency with the motor directly driving the impeller shaft.

The high-efficiency motor surpasses the industry's highest IE5 energy efficiency standard, with high power density.

The motor features an independently integrated electronic control cooling system for higher efficiency.

The fan achieves a static pressure efficiency of over 70%, approximately 5% higher than competing products in the industry.

Low Noise

The special geometric shape of the blades facilitates better gas flow, resulting in lower noise levels.

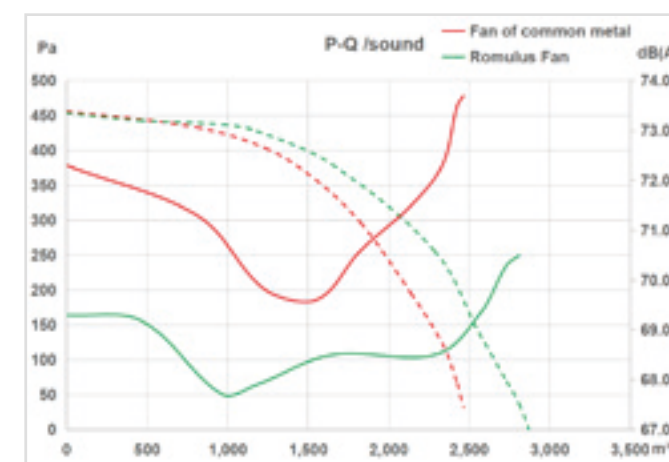
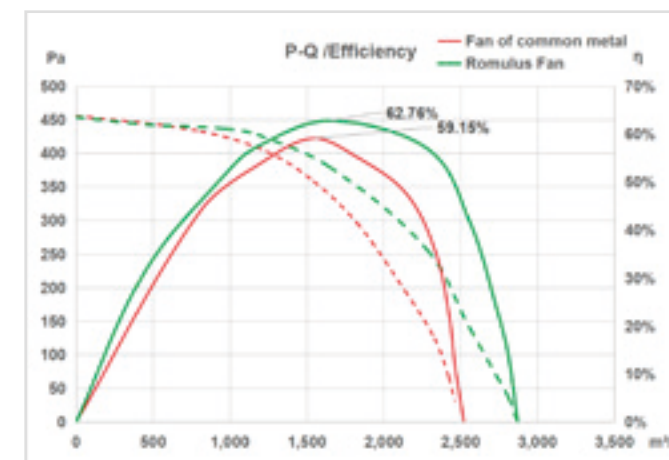
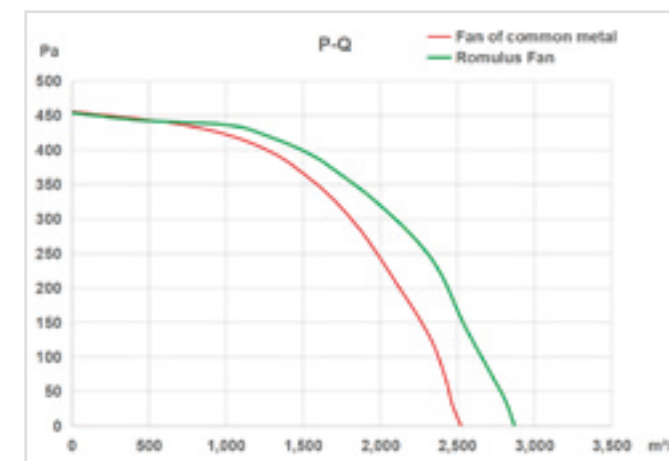
The high-efficiency motor and impeller design reduce noise levels by approximately 2 decibels at common operating points.

Portable Installation

Convenient Installation Integrated installation method that is compact and superior in performance within narrow spaces, saving customers more installation space.



Comparative Advantage



Advantages in Air Volume and Pressure Operating Range

The fluid dynamic design of the impeller, combined with proprietary guide vanes, allows the Revive Air series impellers to achieve a higher air volume at the same speed compared to traditional metal impellers in the market.

Enhanced Efficiency

With the electromagnetic design of the new-generation motor and the CFD-designed impeller, Romulus demonstrates improved efficiency performance.

Reduction in Noise Levels

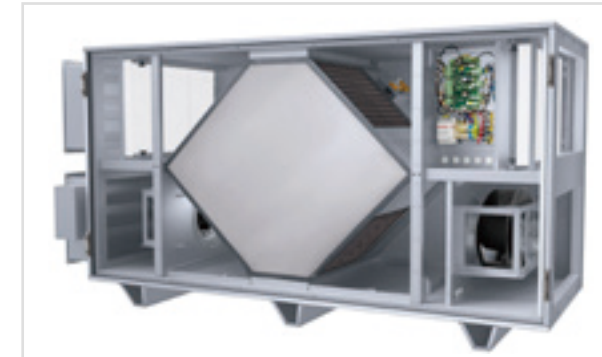
Through the design optimization of motor compatibility, adjustments in impeller installation angles and curvature modifications, as well as the high consistency in the molding of plastic blades, the rotational noise of the impeller has been effectively reduced.

02 Product Applications



Air Purifier

When using air purifiers, we aim for high filtration efficiency, low noise levels, compact design, and reduced energy consumption. The Romulus fan, with its specially designed impeller and EC motor, meets these requirements, providing excellent filtration performance and quiet operation.



AHU

In standard AHU systems, Romulus features an integrated bracket design that simplifies assembly. Its high efficiency and low noise performance make it ideal for both commercial and residential applications. The convenient Modbus control protocol allows precise adjustments of temperature and humidity in the environment.



FFU

For FFU (Fan Filter Unit) applications, which require high efficiency, low noise, and stable group control in medium to low static pressure environments, two highly efficient fan models, 420 and 456, have been specifically developed. These models are tailored to meet the unique demands of FFU systems, ensuring optimal performance in such scenarios.



Silent Duct Fan

For high-end commercial applications, duct fans are required to operate quietly and stably 24/7 and to adjust airflow flexibly based on environmental changes. The Romulus fan meets these needs and enables smart control to optimize energy use.



Mechanical Equipment

In medical devices and laser processing equipment, efficient ventilation and temperature control are crucial, as is fan stability. Romulus's EC motor with multiple protective features and flexible control options effectively meets the stringent requirements of these applications.

03 Production Capacity



Our Persistence

Serving the market with green products through innovative design and continual improvement in manufacturing processes and customer service has always been of prime importance and a key factor in maintaining Seemtek's position at the forefront of the EC motor and fan market. Whether it's one of the standard products or a custom design, Seemtek is ready to working with you for the optimal solutions.

Backward Centrifugal Fan of Romulus



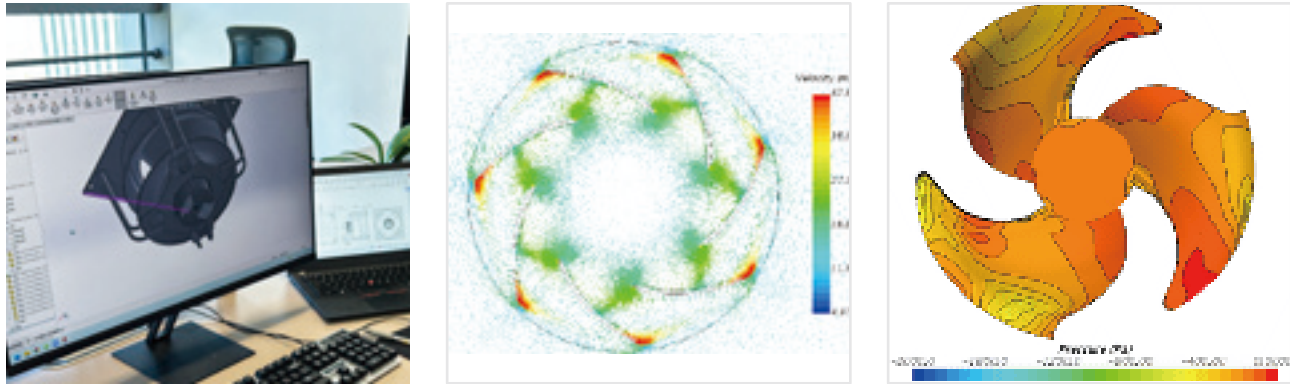
20+
Years
Experience
Verified by 3000+ Clients in 150+ Countries

30+
R&D
Engineers
Driving Cutting-edge Innovation with Expertise

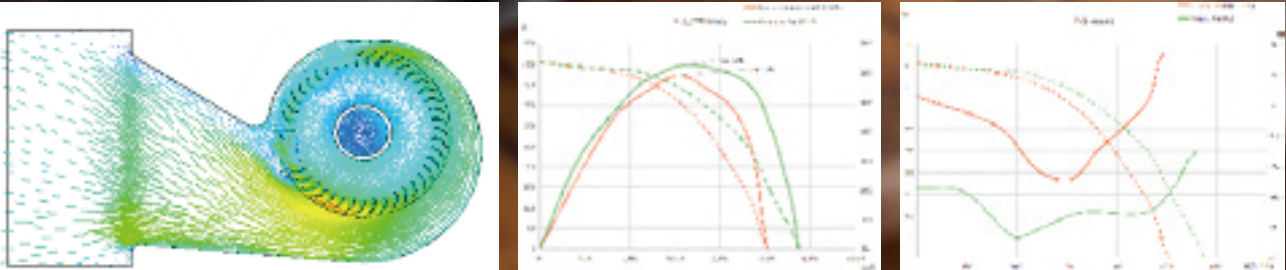
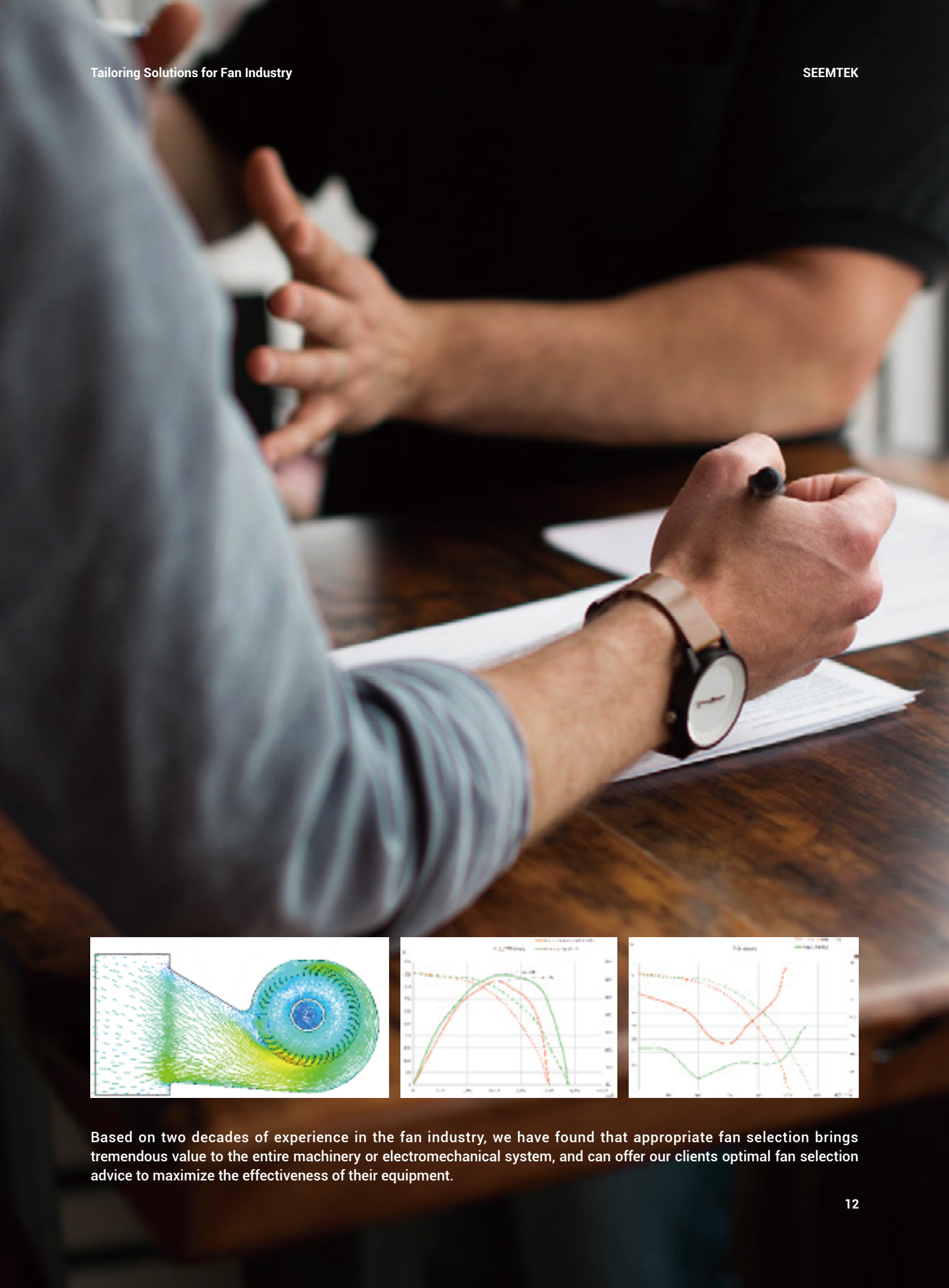
11
Test
Producers
Comprehensive Quality Assurance Through Rigorous Testing

R&D Capability

Through 20 years focused on electromagnetic technology and fluid design, Seemtek has always firmly believed that research and innovation are crucial to its competitiveness, which is why it focuses on constantly building and enhancing its R&D division.



We now have an R&D team of over 30 members. From initial demands to final results, our R&D team always remain by our customers' side, helping them find the most effective solutions to elevate the value of their equipment.



Based on two decades of experience in the fan industry, we have found that appropriate fan selection brings tremendous value to the entire machinery or electromechanical system, and can offer our clients optimal fan selection advice to maximize the effectiveness of their equipment.

Reliability Testing



Noise Test

This test verifies the low noise performance of the fan at high speeds and high static pressure, analyzing noise through FFT noise spectrum analysis. The optimized flow path and blade design effectively reduce noise levels, improving the overall comfort of the operating environment. It minimizes the transmission of low-frequency noise over long distances in ventilation ducts.

Airflow Test

This test measures the airflow output and fan efficiency under different speeds and static pressure resistance. The optimized impeller design and three-dimensional blade structure of the Romulus fan ensure efficient airflow output and adaptability to various working conditions.



Constant Temperature & Humidity Test

This test validates the fan's performance stability in environments with varying temperature and humidity, particularly in corrosion and moisture-resistant conditions. A 21-day continuous operation at 75°C and 95% humidity is used to verify the fan's long-term stability.



Salt Mist Test

Salt mist test assesses the corrosion resistance of the Romulus fan, especially under C5-grade corrosion conditions. It ensures long-term performance in salty environments, providing protective measures for aluminum-magnesium alloy impellers against corrosion.



Temperature Shock Test

This test evaluates the operational stability of the fan under rapid temperature changes. The high-temperature materials and precise control system of the Romulus fan ensure that performance is not affected by thermal expansion and contraction under extreme temperature variations.



High-Speed Lifespan Test

This test focuses on verifying the strength and stability of the Romulus fan's impeller under long-term high-speed operation and ensures that the impeller remains free from deformation or damage under high-speed conditions, ensuring the fan's safety and reliability, especially in applications requiring high-efficiency airflow output.



Outdoor Lifespan Test

This test validates the fan's durability in harsh outdoor environments. The Romulus fan's IP55/IP56 protection rating, along with its high-temperature and corrosion-resistant design, ensures reliable long-term operation outdoors.



Indoor Lifespan Test

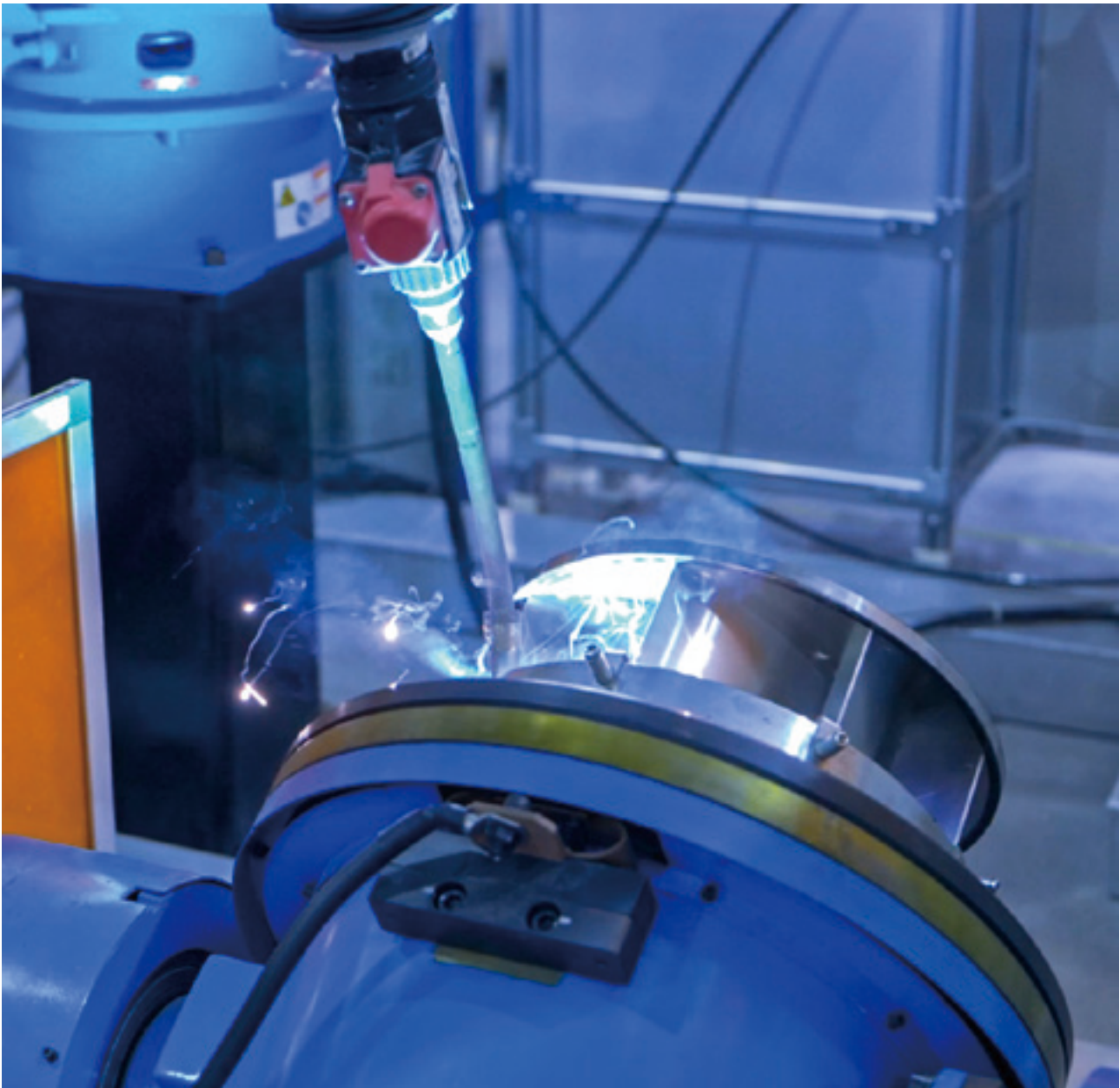
This test evaluates the long-term lifespan of the fan in standard indoor environments, ensuring that the Romulus fan has a long life with low maintenance requirements. The test is particularly focused on its performance in common applications such as AHU and FFU systems.



High Temperature Test

This test validates the reliable operation of the Romulus fan in high-temperature environments, especially under high static pressure and efficient electromagnetic design conditions. It ensures the stability of the impeller material at extreme temperatures, extending the fan's lifespan.

Factory Workshop



Seemtek currently has various fully automated production lines in use, producing millions of high-performance axial fans and centrifugal fans every year. These production lines ensure the highest quality and efficiency, meeting the rigorous demands of our global customers. With the expertise and dedication of our skilled workforce and advanced manufacturing techniques allows us to deliver superior products that set industry standards for performance and reliability.



04

Product Catalog

The Detailed Parameters of Romulus

Superior product performance

With professional aerodynamics and electromagnetic design, coupled with industry-proven control and drive technologies, we ensure that each product achieves excellent performance.



Part Numbering System

<u>B</u>	<u>3</u>	<u>E</u>	-	<u>2A</u>	<u>190</u>	-	<u>S0</u>	-	<u>00</u>
1	2	3		4	5		6		7

1: Fan Type

B: Backward Curved Centrifugal Fan
 K: Centrifugal Fan with Bracket
 F: Forward Curved Centrifugal Fan
 D: Dual Inlet Blower Fan
 S: Single Inlet Blower Fan
 A: Axial Fan
 G: Axial Fan with Guard Grille
 W: Axial Fan with Wall Ring
 M: Motor

2: Number of Cores(EC) / Number of Poles(AC)

EC: 1-core and 3-core
 AC: 2, 4, 6, 8 and 12 poles

3: Type of Motor

E: EC brushless motor
 D: DC brushless motor
 A: AC single phase induction motor

4: Input Voltage

12: 12 VDC
 24: 24 VDC
 48: 48 VDC
 85: 85 VDC
 1D: 110 VDC
 3D: 310 VDC
 1A: 115 VAC
 2A: 230 VAC
 3A: 380 VAC

5: Diameter

190: impeller diameter in mm

6: Key for Electrical Design

7: Key for Mechanical Design



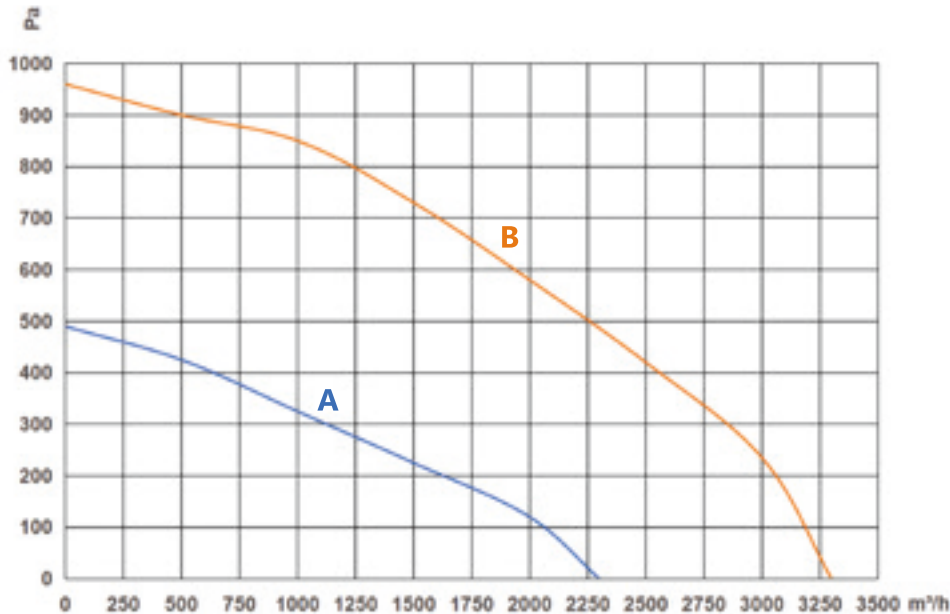
Impeller Material: Plastic
Direction of Rotation: Clockwise, seen on rotor
Speed Control: 0-10VDC/PWM
Type of Protection: Ip44/54
Operation Mode: S1
Bearing: Ball Bearing
Ambient Temperature: - 25 °C ~ +60 °C



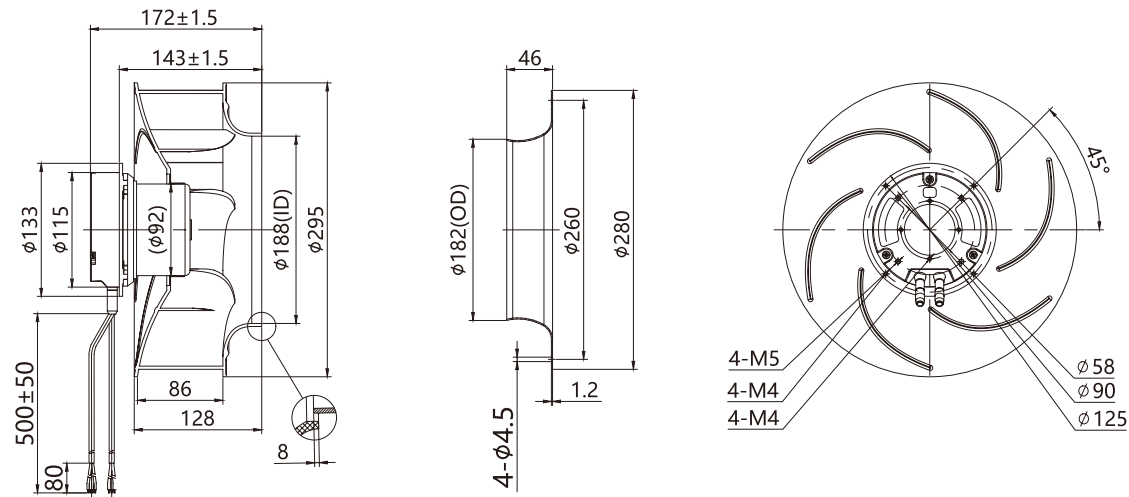
Technical Parameters

Model No.	Rated Voltage VAC	Current A	Input Power W	Speed RPM	Air Flow m³/h	Air Pressure Pa	Insulation	Noise Level dBA	Curve
B3E-2A280-K9-00	230	1.40	158	1860	2300	490	Class B	68	A
K3E-2A280-K9-00	230	1.40	158	1860	2300	490	Class B	68	
B3E-2A280-K2-01	230	2.20	480	2650	3300	975	Class F	72	B
K3E-2A280-K2-01	230	2.20	480	2650	3300	975	Class F	72	

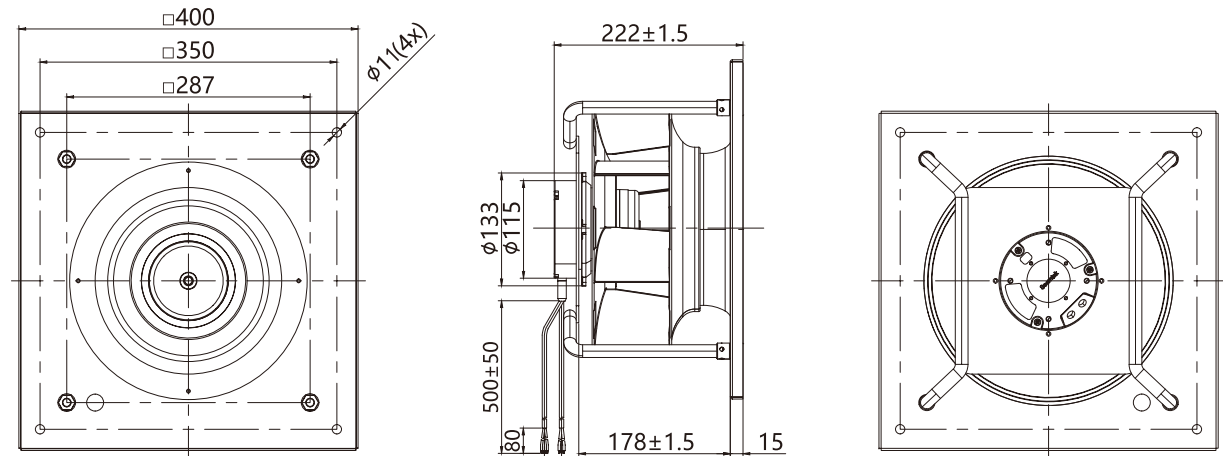
Performance Curve



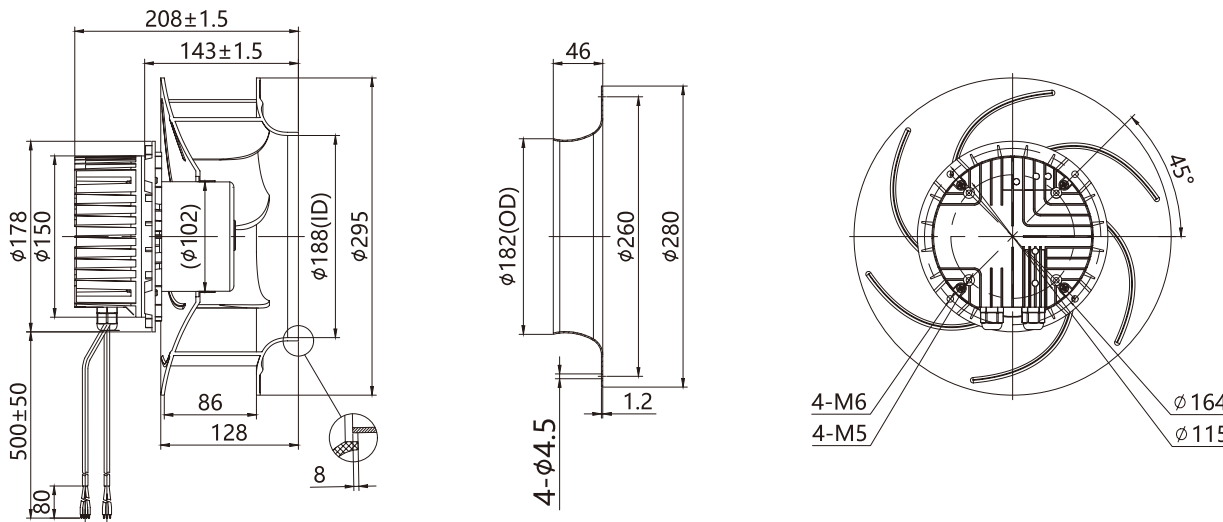
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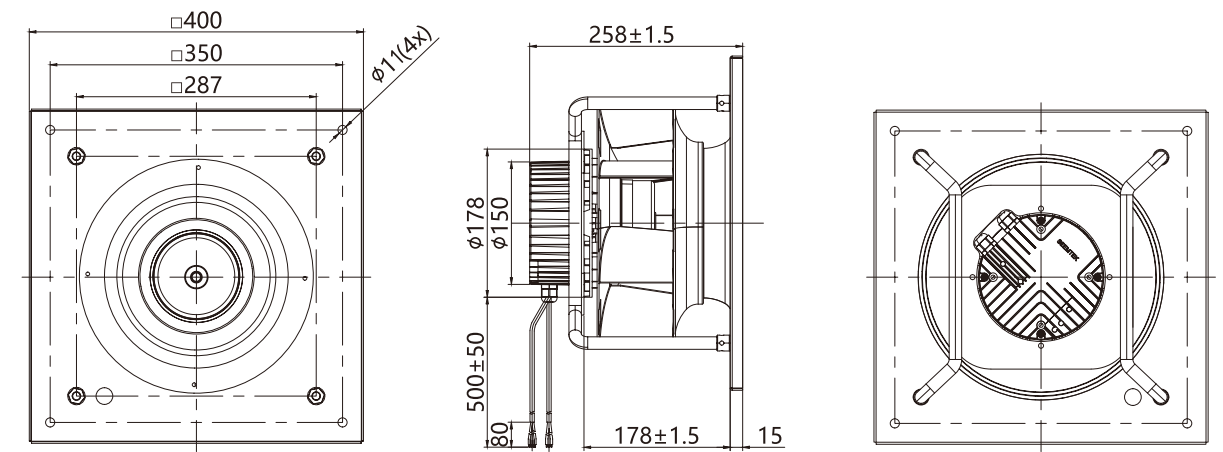
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B3E-2A280-K2-01



K3E-2A280-K2-01



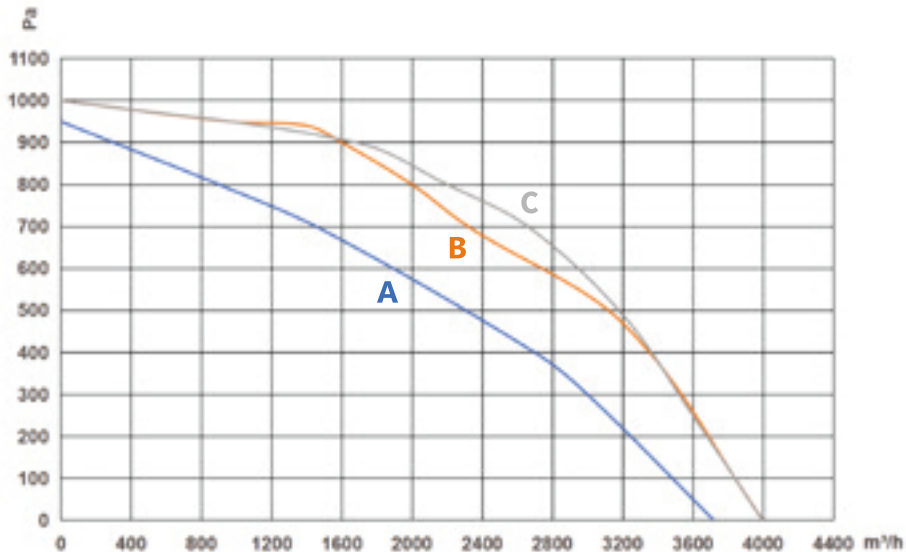


Impeller Material: Plastic
Direction of Rotation: Clockwise, seen on rotor
Speed Control: 0-10VDC/PWM
Type of Protection: IP54
Operation Mode: S1
Bearing: Ball Bearing
Ambient Temperature: - 25 °C ~ +60 °C

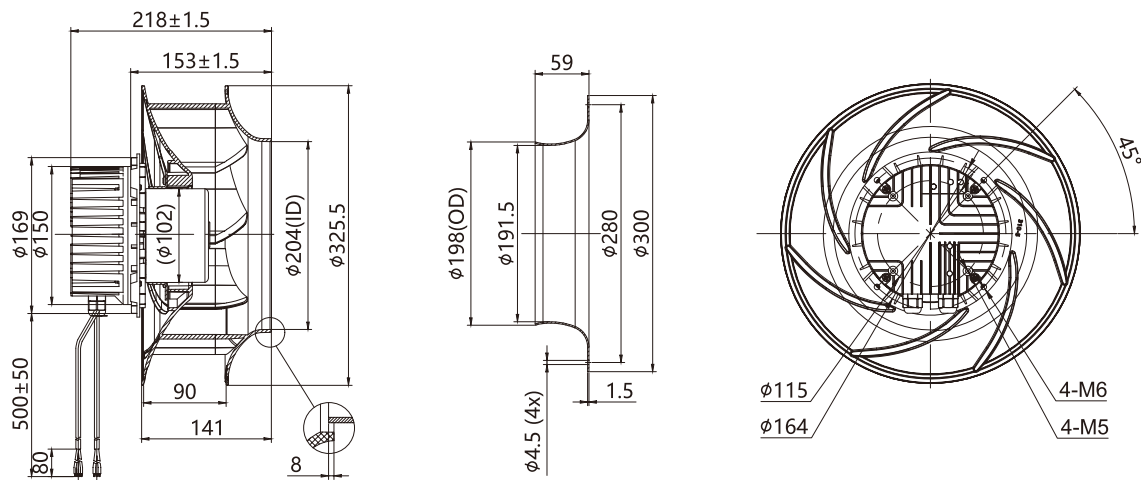


Technical Parameters									
Model No.	Rated Voltage VAC	Current A	Input Power W	Speed RPM	Air Flow m³/h	Air Pressure Pa	Insulation	Noise Level dBA	Curve
B3E-2A310-K2-00	230	2.20	480	2310	3721	950	Class F	72	A
K3E-2A310-K2-00	230	2.20	480	2310	3721	950	Class F	72	
B3E-2A310-K2-01	230	3.25	730	2650	4000	1000	Class F	78	B
K3E-2A310-K2-01	230	3.25	730	2650	4000	1000	Class F	78	
B3E-3A310-K2-00	380	1.25	770	2650	4000	1000	Class F	78	C
K3E-3A310-K2-00	380	1.25	770	2650	4000	1000	Class F	78	

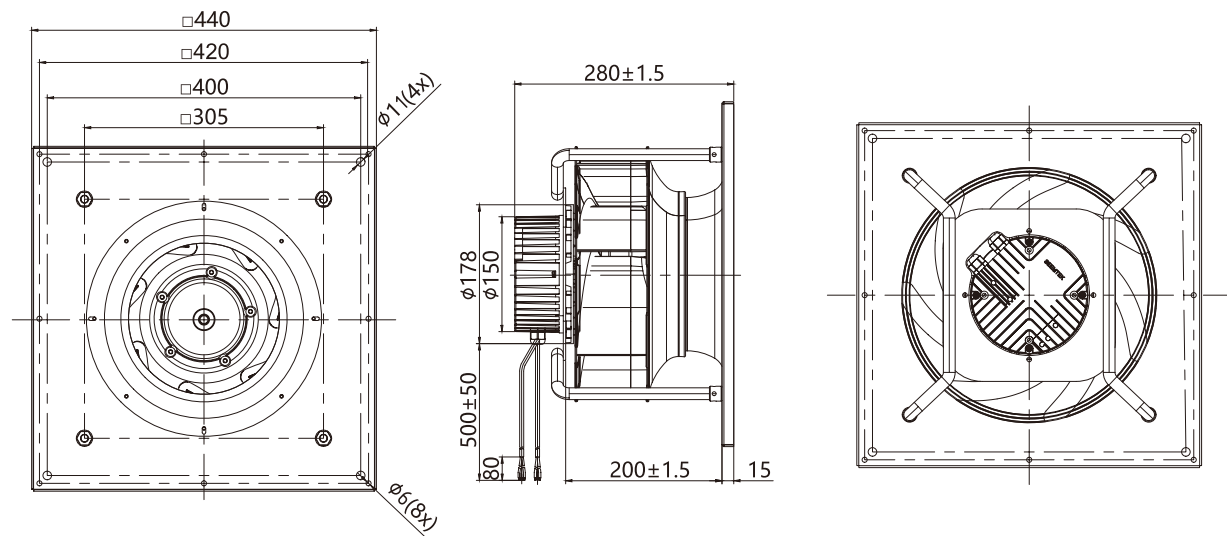
Performance Curve



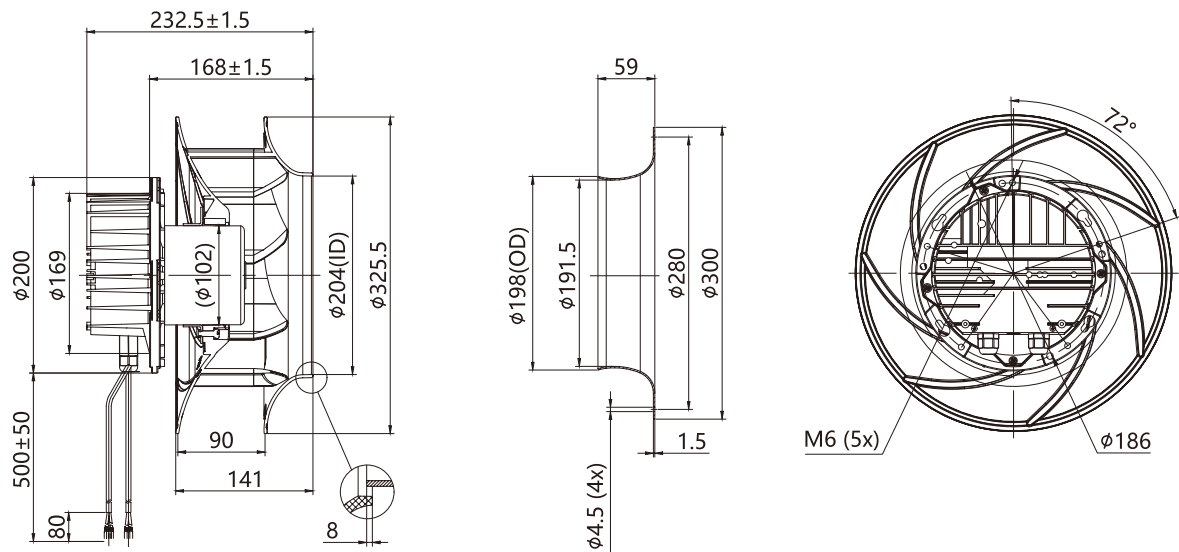
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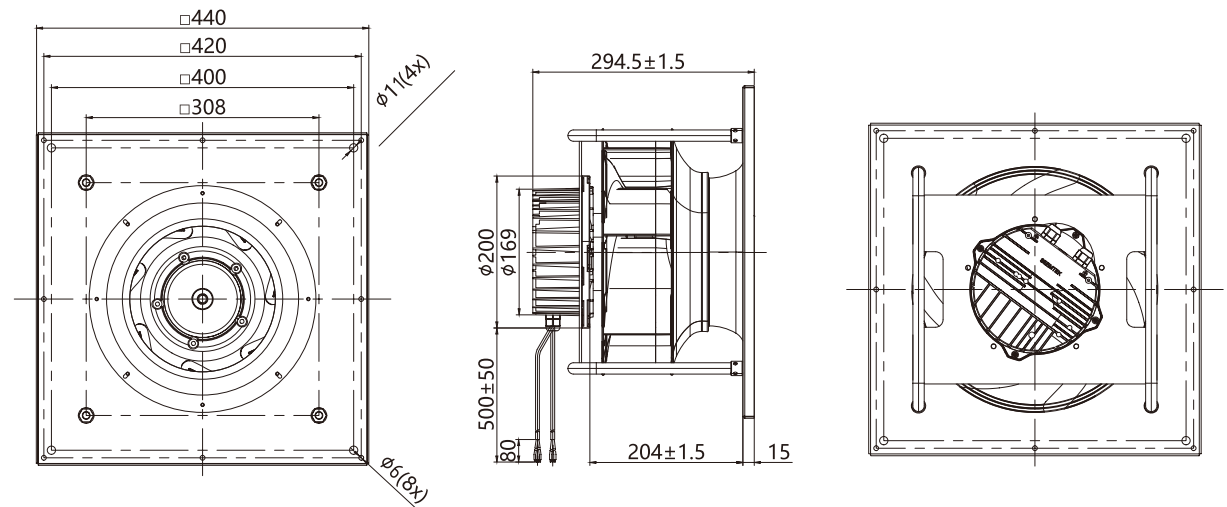
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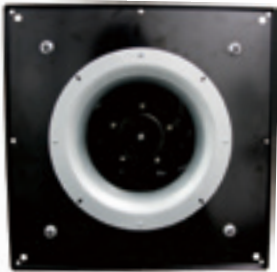


B3E-2A310-K2-01 & B3E-3A310-K2-00



K3E-2A310-K2-01 & K3E-3A310-K2-00

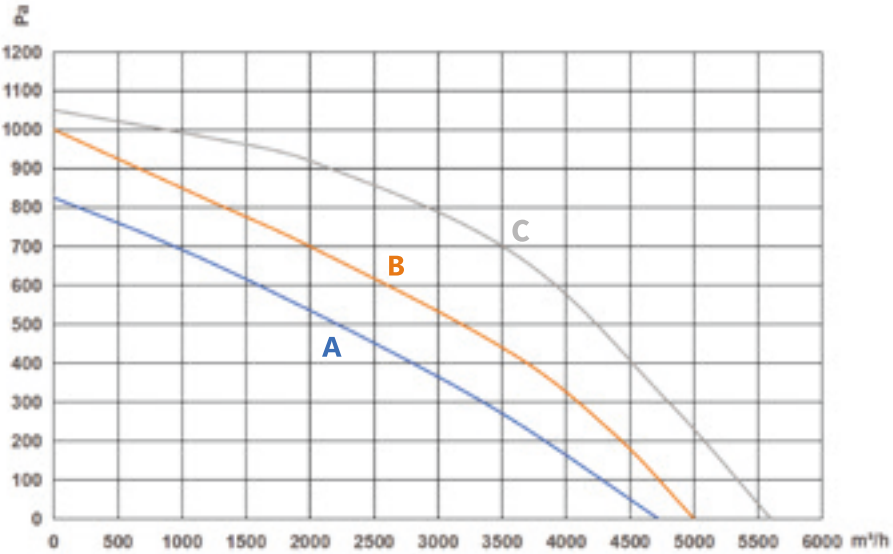




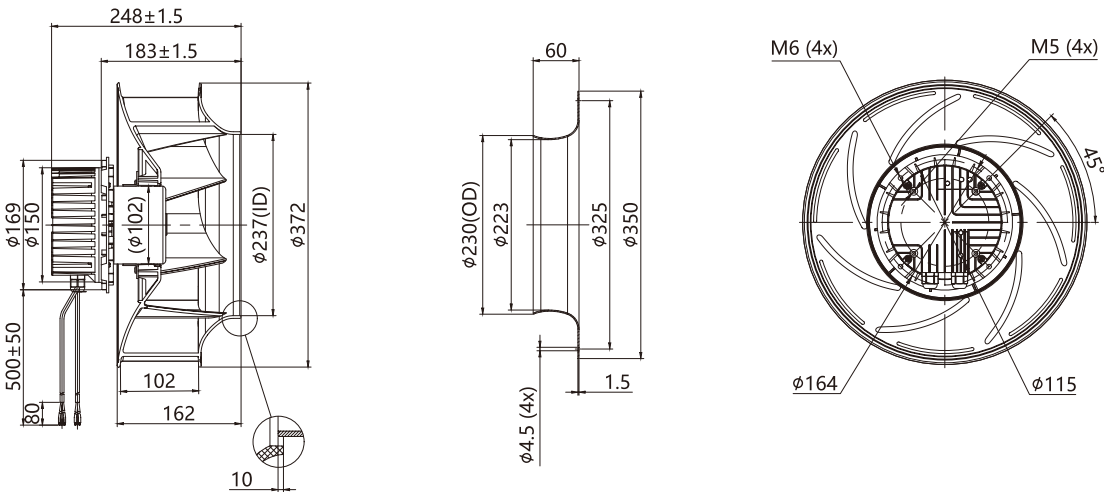
Impeller Material: Plastic
Direction of Rotation: Clockwise, seen on rotor
Speed Control: 0-10VDC/PWM
Type of Protection: IP54
Operation Mode: S1
Bearing: Ball Bearing
Ambient Temperature: - 25 °C ~ +60 °C

Technical Parameters									
Model No.	Rated Voltage VAC	Current A	Input Power W	Speed RPM	Air Flow m³/h	Air Pressure Pa	Insulation	Noise Level dBA	Curve
B3E-2A355-K2-00	230	2.20	480	1800	4720	825	Class F	73	A
K3E-2A355-K2-00	230	2.20	480	1800	4720	825	Class F	73	
B3E-2A355-K2-01	230	3.30	730	2045	5000	1000	Class F	77	B
K3E-2A355-K2-01	230	3.30	730	2045	5000	1000	Class F	77	
B3E-3A355-K3-00	380	1.80	1130	2350	5600	1050	Class F	83	C
K3E-3A355-K3-00	380	1.80	1130	2350	5600	1050	Class F	83	

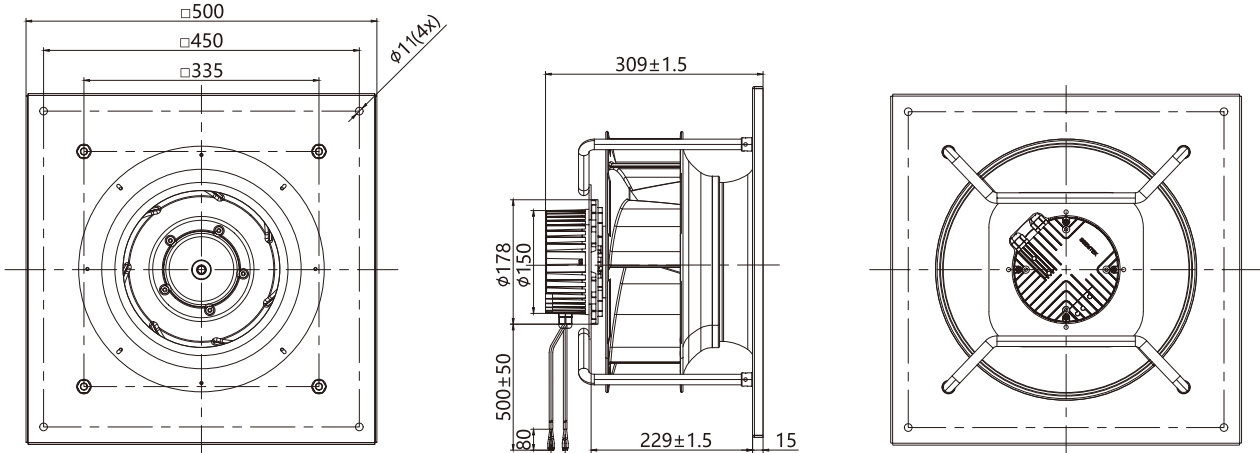
Performance Curve



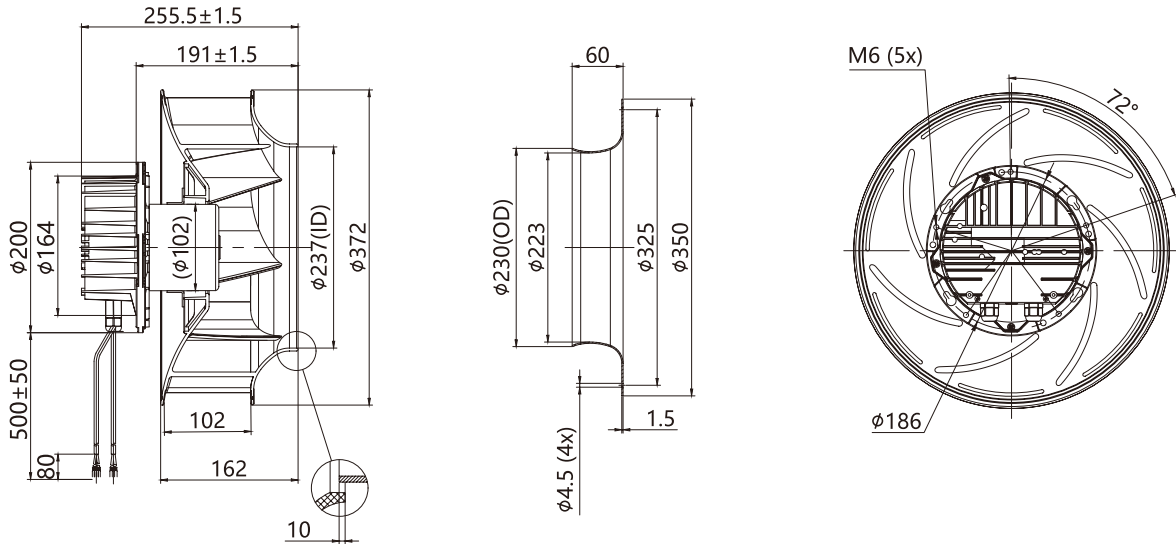
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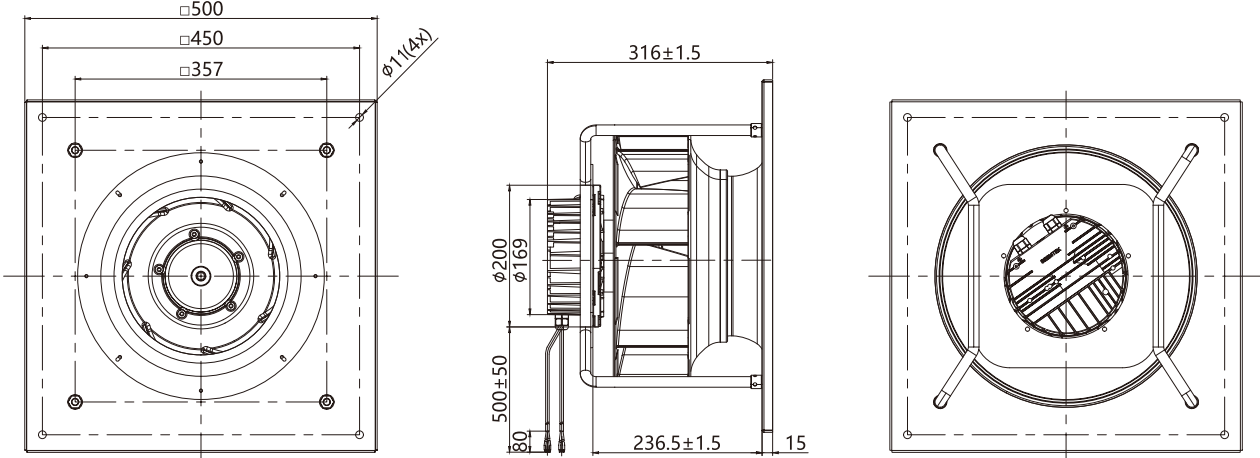
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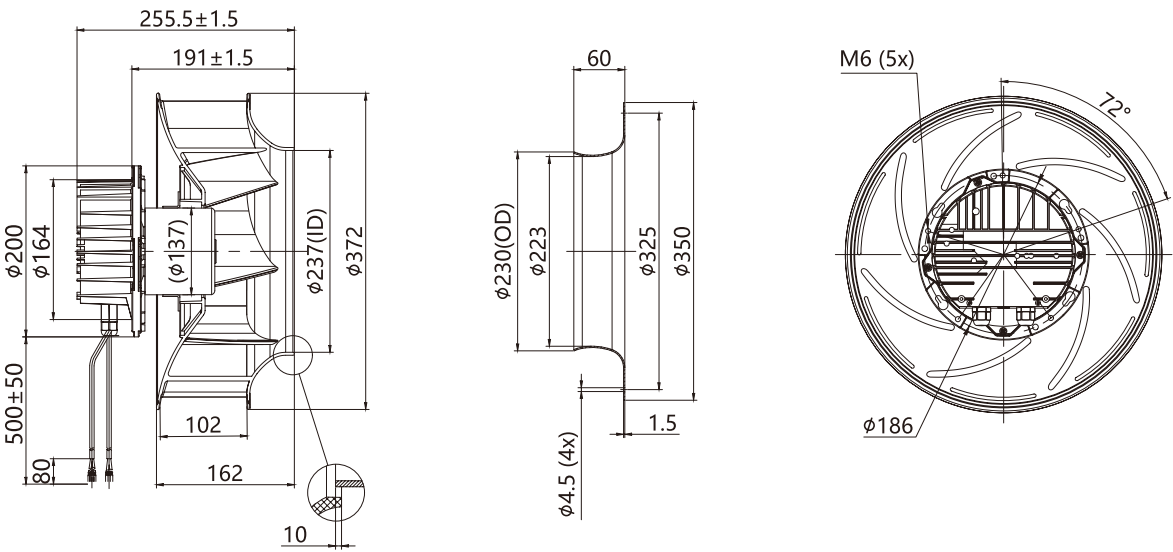
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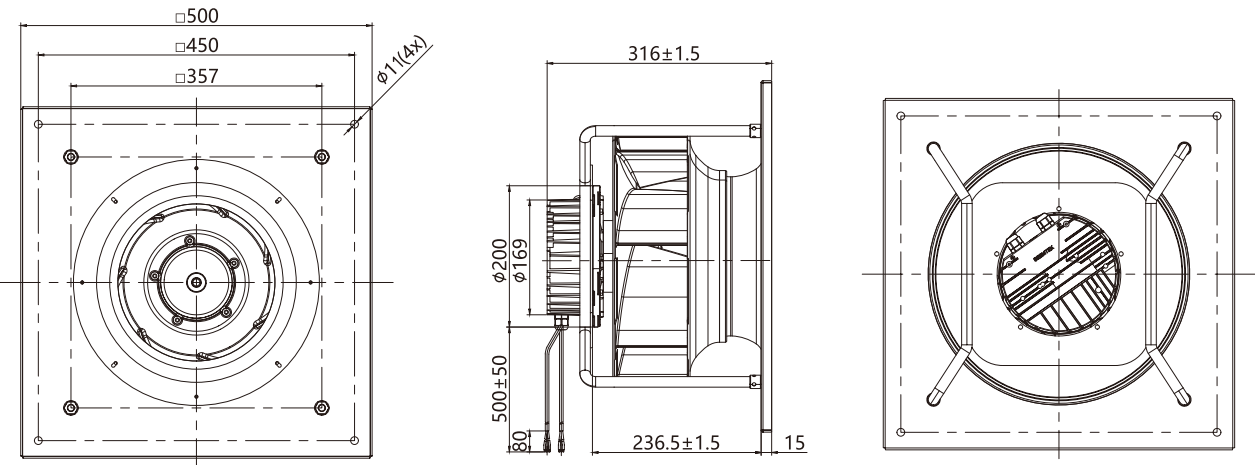
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B3E-3A355-K3-00



K3E-3A355-K3-00



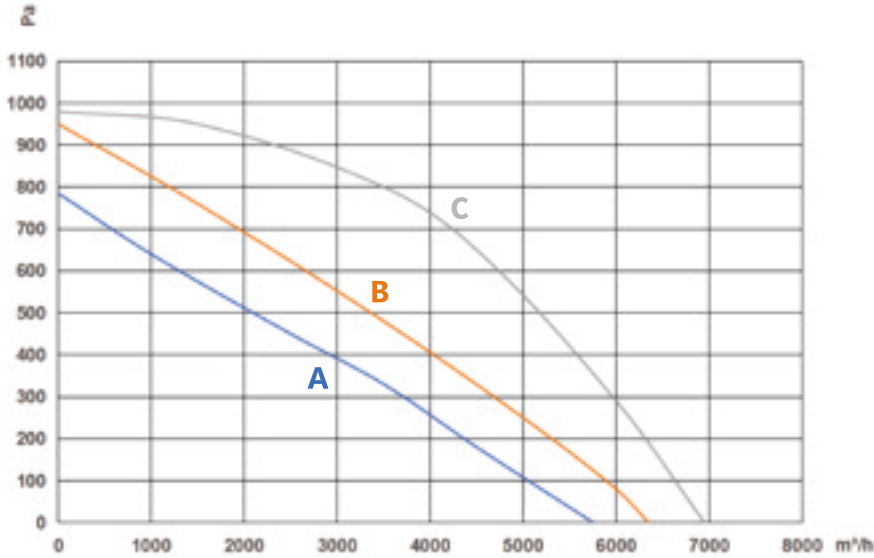
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Direction of Rotation: Clockwise, seen on rotor
Speed Control: 0-10VDC/PWM
Type of Protection: IP54
Operation Mode: S1
Bearing: Ball Bearing
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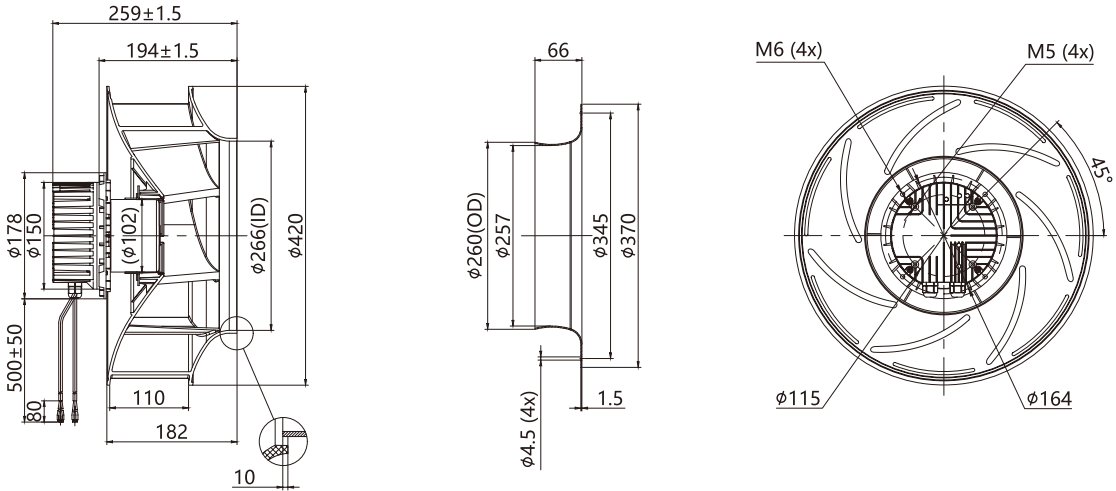
Technical Parameters

Model No.	Rated Voltage VAC	Current A	Input Power W	Speed RPM	Air Flow m³/h	Air Pressure Pa	Insulation	Noise Level dBA	Curve
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K3E-2A400-K2-00	230	2.20	480	1450	5755	785	Class F	72	A
B3E-2A400-K2-01	230	3.30	730	1650	6350	950	Class F	76	B
K3E-2A400-K2-01	230	3.30	730	1650	6350	950	Class F	76	B
B3E-3A400-K3-00	380	2.10	1300	2010	6950	980	Class F	83	C
K3E-3A400-K3-00	380	2.10	1300	2010	6950	980	Class F	83	C

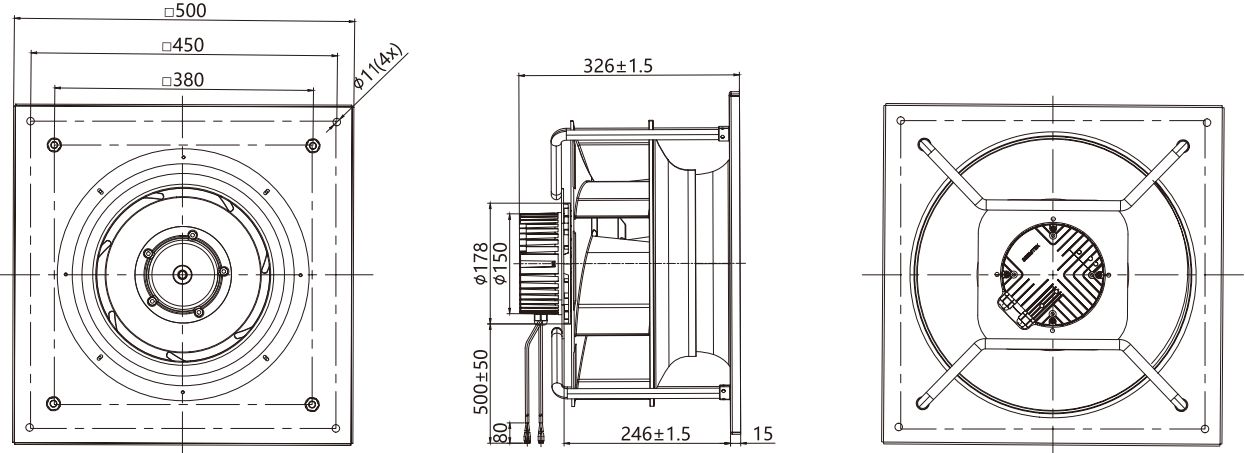
Performance Curve



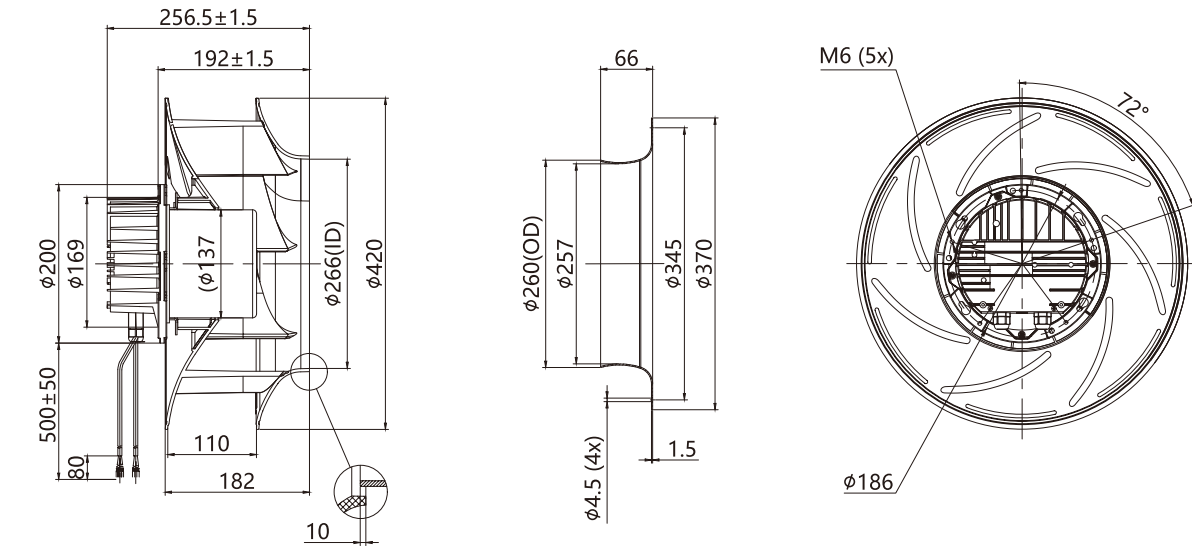
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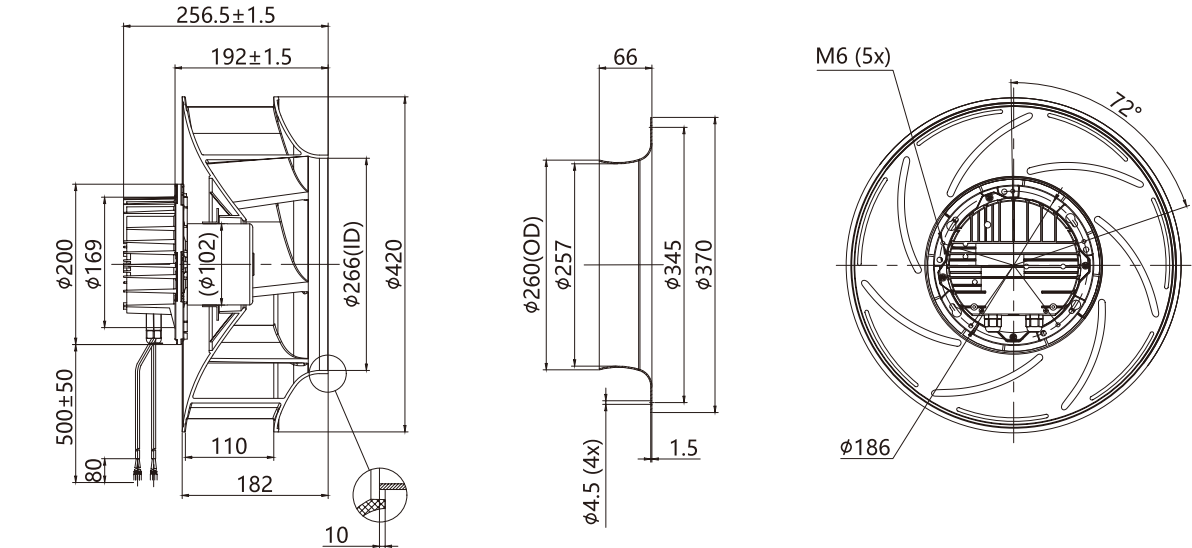
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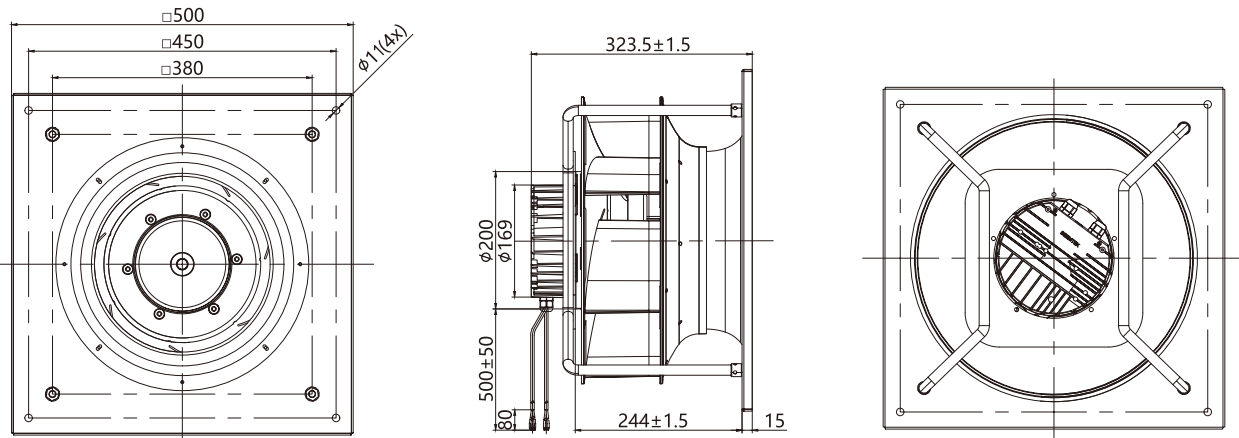
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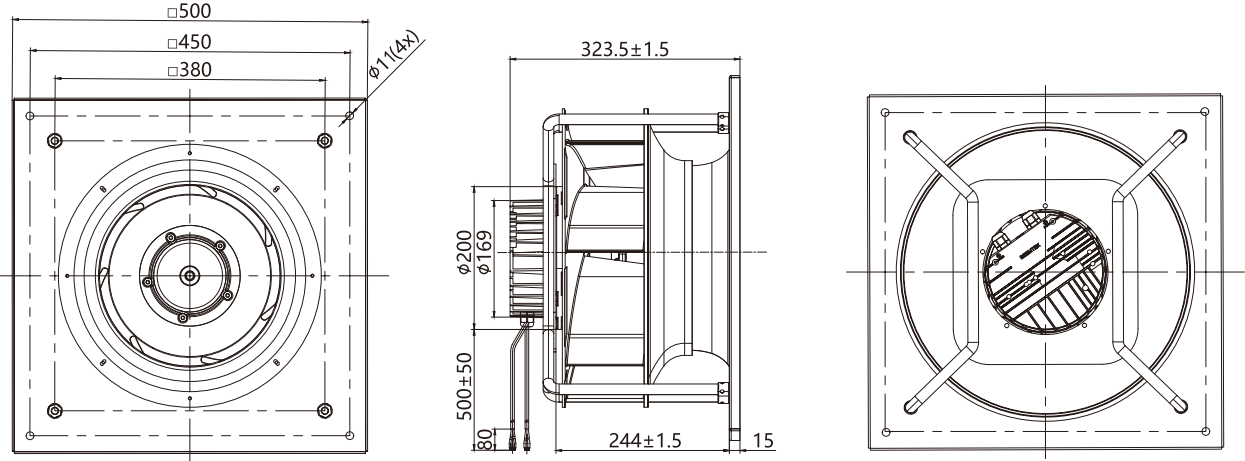
B3E-2A400-K2-01



K3E-3A400-K3-00



K3E-2A400-K2-01



SMT49C2410E-32

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